

Effect of Constraining a Limited Number of Slant Ionospheric Corrections in PPP and an Improved Partial Tight-Constraining Algorithm

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With external ionospheric corrections, precise point positioning solutions can be augmented with proper constraints, enabling fast convergence and high-precision positioning performance. However, there are two main issues in user algorithms. First, the contribution of the number of applied ionospheric corrections to positioning performance is not clear to users. For instance, in low-cost sensors where computation resources are limited, applying too many corrections (e.g., tens of corrections) from the server will possibly cause a slow response. Second, the determination of constraining strength for corrections from different satellites remains challenging. Therefore, this article evaluates the effect of the number of ionospheric corrections on positioning errors and convergence time, and proposes an improved tight-constraining algorithm for partially reliable corrections. Static and simulated kinematic positioning results confirm that the more corrections applied, the better the positioning performance. However, the positioning accuracy only has limited improvements when several corrections are already applied, especially in the north direction. The proposed method reduces static positioning root-mean-square errors by 43.4%, 26.9%, and 32.1% in the east, north, and up directions, respectively, compared to solutions without constraint. In simulated kinematic results, the convergence time is further shortened by 7.7%, 4.8%, and 2.5% for the east, north, and up directions, respectively, compared to cases where all corrections are applied.

I. INTRODUCTION

Precise point positioning real-time kinematic (PPP-RTK) technology has gained significant attention in recent years because of its capability of providing high-precision global navigation satellite system (GNSS) positioning solutions within a short period of time [1], [2], [3], [4]. A relatively long convergence time remains the main barrier to GNSS practical applications. It has been validated that, with low Earth orbit (LEO) satellite augmentation and atmospheric corrections, convergence time can be reduced [5], [6], [7]. One of the most critical corrections in PPP-RTK is the precise ionospheric information, as it largely affects the precision, convergence time, and ambiguity resolution (AR) performance [8], [9]. Therefore, properly applying these ionospheric corrections is of vital importance in achieving satisfactory positioning performance that meets diverse user demands.

A. Related Work

Institutions such as international GNSS service provide global ionosphere map (GIM) products to users, through which the ionospheric corrections at a location can be calculated with the interpolated vertical total electron content and the corresponding ionospheric mapping function. Although positioning performance can be improved with such vertical ionospheric corrections [10], [11], the precision of GIM and other vertical ionospheric models is approximately 2–7 total electron content unit (TECU) [12], [13], [14], which cannot satisfy higher-precision GNSS requirements [15]. Moreover, the precision of calculated slant ionospheric corrections using GIM is further limited to the accuracy of different mapping functions, especially when the satellite elevation angle is low [16], [17]. Consequently, in small regions, slant ionospheric corrections are typically generated and directly applied to achieve fast convergence and high-precision positioning [18], [19].

Research on slant ionospheric corrections mainly focuses on two aspects: correction modeling and uncertainty determination. In terms of slant ionospheric correction modeling, polynomial functions of different orders are widely used to fit the coefficients of correction models for each satellite. It was reported that the model precision can be improved by using modified spherical harmonic and generalized trigonometric series functions [20]. Similarly, Li et al. [21] proposed slant ionospheric modeling with a satellite-based ionospheric model and achieved a correction bias within 1.0 TECU. Compared to modeling the slant ionosphere with mathematical functions, slant ionosphere information estimated from nearby server stations can also be directly applied to users, and it performs well in small regions due to the similar propagation path of the GNSS signals [22]. More recently, atmospheric corrections derived from PPP-AR are applied in factor graph-based PPP-RTK solutions, results showed that the AR performance and positioning accuracy are significantly improved [23].

The corrections serve as components in the function model, whereas the corresponding stochastic model, i.e., uncertainty information, is also important in the filter. When only an optimal *a priori* variance is determined for the ionospheric correction, the positioning performance can be improved [24]. Wang et al. [25] presented a new interpolation method and its use in calculating the uncertainties of interpolated corrections; the results showed that the fixing rate of AR is higher than 90%. The intersatellite cross-verification error function was proposed by Li et al. [26], and the results demonstrated that with the improved uncertainty determination function, the convergence time was shortened by 10%–60% compared with fixed ionospheric uncertainties of 1, 8, and 30 cm. Moreover, the uncertainty grids method was also proposed and improved [27]. Cui et al. [28] generated uncertainty grids together with the corrections and applied them to user algorithms, showing that the convergence time can be sped up by 42.1% and 28.8% for horizontal and vertical components, respectively. To achieve quality monitoring of atmospheric corrections, Huang et al. [29] proposed leave-one-out cross-validation and improved positioning accuracy of more than 20%. Zhang et al. [30] incorporated warning information with ionospheric corrections and applied the broadcast information in network real-time kinematic (RTK) to improve positioning performance during ionospheric activities. In conclusion, the correction uncertainty information, when accurately represented and constrained by the user, can significantly improve positioning performance. Currently, uncertainty information is generally broadcast by the server, and few studies focus on methods to determine the uncertainty and select the proper constraining strength for the user.

B. Motivations and Contributions

In most research, all available corrections are applied on the user side. Considering that there could potentially

be more than 30 satellites in view at a location for a multi-GNSS case, which thereby raises the following questions that: 1) Is it necessary to apply all the corrections for users and 2) What is the quantitative contribution of increasing the number of corrections? It was simulated and validated that even with one wide-lane ambiguity fixed, the single-epoch positioning performance is improved [31]. However, the contribution of constraining limited ionospheric corrections in positioning performance has not yet been investigated.

Given the above-mentioned limitations of current research in ionosphere-augmented GNSS solutions, this study evaluates the effect of constraining a limited number of slant ionospheric corrections in PPP-RTK, and a novel partial tight-constraining strategy is proposed as well, to adapt to the selected reliable corrections. The key contributions are outlined as follows.

- 1) The effect of the number of slant ionospheric corrections on PPP-RTK positioning performance and convergence time is quantitatively analyzed for the first time. The results, from a unique perspective, provide answers to the question that how many corrections are needed to meet different user demands.
- 2) The server correction quality indices, including the maximum difference among slant ionospheric corrections, corresponding standard deviations, comparison between filtered variance and *a priori* constraint strength, and the number of continuously processed epochs for a specific ionospheric parameter, are detailed, introduced, and analyzed. The results provide a valuable reference for adjusting constraining strength.
- 3) Unlike existing research, where uncertainty information is broadcast by the server, this study proposes an adaptive constraining strength strategy for partial ionospheric corrections from the user's perspective. In this approach, the constraining strength of qualified corrections is either loosened or tightened, providing a more accurate and reliable positioning solution, along with faster convergence compared to conventional methods.

II. METHODOLOGY FOR CONVENTIONAL PPP METHOD WITH STEC CONSTRAINING

This section briefly introduces the conventional precise point positioning mathematical model, as well as the method for interpolating the STEC corrections and the newly proposed partial tight-constraining algorithm.

A. Precise Point Positioning Model

The undifferenced and uncombined (UDUC) model is usually used in PPP-RTK, as it contains the parameters necessary for constraining. The j th frequency GNSS pseudorange (P) and carrier phase (L) measurements for satellite

s are as follows:

$$\begin{cases} P_j^s = \rho^s + c(dt_r - dt^s) + \gamma_j I_1^s + T^s + b_j^s + b_{r,j} + \varepsilon_j^s \\ L_j^s = \rho^s + c(dt_r - dt^s) - \gamma_j I_1^s + T^s + \lambda_j (N_j^s + B_j^s + B_{r,j}) + \xi_j^s \end{cases} \quad (1)$$

where both measurements are in units of meters. ρ is the satellite–receiver geometrical range, and c is the speed of light. dt_r and dt^s are the receiver and satellite clock offset, respectively. γ_j is the ionospheric frequency coefficient on the j th frequency, and $\gamma_j = (f_1/f_j)^2$. I_1^s is the slant ionospheric effect on the first frequency, and T^s is the slant tropospheric delay. λ_j is the wavelength, and N_j^s is the integer ambiguity. b_j^s , $b_{r,j}$, B_j^s , and $B_{r,j}$ are code and phase hardware biases for the satellite and receiver, respectively. ε_j^s and ξ_j^s are code and phase observation noise.

With precise satellite orbit and clock products from analysis centers, after applying proper error model corrections, the linearized dual-frequency estimable PPP equation can be presented as follows:

$$\begin{bmatrix} \hat{P}_j^s \\ \hat{L}_j^s \end{bmatrix} = \begin{bmatrix} e_x^s & e_y^s & e_z^s & 1 & mf_w^s & \gamma_j & 0 \\ e_x^s & e_y^s & e_z^s & 1 & mf_w^s & -\gamma_j & \lambda_j \end{bmatrix} \begin{bmatrix} dX & dY & dZ & \tilde{dt}_r & ZWD & \tilde{I}_1^s & \tilde{N}_j^s \end{bmatrix}^T \quad (2)$$

where $\hat{P}$ and $\hat{L}$ are the measurement misclosures, e_x^s , e_y^s , and e_z^s are satellite–receiver line-of-sight unit direction vectors, mf_w^s is the wet mapping function of tropospheric zenith wet delays (ZWD). The tilde markers infer that the parameters have biases. dX , dY , and dZ are the corrections to the given *a priori* initial approximate coordinates, $\tilde{dt}_r$ is a combination of receiver clock and receiver hardware delays, ZWD is the tropospheric ZWD. $\tilde{I}_1^s$ denotes a combination of ionospheric delays on the first frequency and satellite differential code bias (DCB). $\tilde{N}$ is the float ambiguity, which absorbed the hardware phase delays and unmodeled errors. The corrected model errors in $\hat{P}$ and $\hat{L}$ are phase center offset and phase center variation [32], phase windup [33], tidal loading, and relativistic effect [34].

B. Correction Interpolation and Partial Tight-Constraining Method

When there are nearby reference stations available (e.g., three server stations in this study), the slant total electron content (STEC) estimated from these stations can be applied as corrections to the user, mitigating the effect of ionospheric refraction on user positioning performance. The criteria for selecting the server stations are that the user station should be close to the center of the triangulation network formed by three server stations. Normally, postprocessed PPP AR are conducted in generating corrections, and the user STEC correction value can be interpolated as follows:

$$\text{stec}_{\text{corr}}^s = \frac{\sum_1^3 (\text{stec}_i^s / \text{dis}_i^2)}{\sum_1^3 (1 / \text{dis}_i^2)} \quad (3)$$

where stec_i^s is the slant ionospheric estimates for satellite s and server station i , and dis_i is the distance from the i th server station to the user. To avoid the effect of receiver

DCB, which is coupled with STEC, between-satellite differenced STEC corrections with respect to reference satellite k ($\tilde{I}^1 - \tilde{I}^k, \dots, \tilde{I}^s - \tilde{I}^k$) are applied as virtual observations and appended to the raw UDUC PPP mathematical model. Hence, with ς representing the residuals, the virtual observation is as follows:

$$\begin{bmatrix} \tilde{I}^1 - \tilde{I}^k \\ \vdots \\ \tilde{I}^s - \tilde{I}^k \end{bmatrix} = \begin{bmatrix} 1 & \cdots & -1 & \cdots & 0 \\ & & & \ddots & \\ 0 & \cdots & -1 & \cdots & 1 \end{bmatrix} \begin{bmatrix} I^1 \\ \vdots \\ I^k \\ \vdots \\ I^s \end{bmatrix} + \begin{bmatrix} \varsigma I^1 - \varsigma I^k \\ \vdots \\ \varsigma I^s - \varsigma I^k \end{bmatrix}. \quad (4)$$

The corresponding constraining strength, i.e., the noise of virtual observations, can first be calculated based on the error propagation law. The following key indices are used to further loosen or tighten the constraints; the detailed analysis can be found in the following sections.

- 1) Maximum difference (also referred to as range) among STEC corrections. There are three STEC corrections available for the same satellite per epoch, which, in theory, should be similar within small regions. The maximum difference among these three STEC corrections reflects the degree of dispersion. When the range is small, the corresponding interpolated STEC correction is expected to be close to the “true” value; therefore, the constraining strength should be tightened. Conversely, when the range is large, the constraining strength should be loosened. In this study, empirical thresholds of 0.1 and 0.6 m are used: tightening the constraint by 100 times for the former and loosening it by 10 times for the latter. Related analysis can be found in Section IV-B.
- 2) Standard deviation of STEC corrections. The consistency of server STEC corrections can also be characterized by the standard deviation of these corrections. Once the standard deviation is smaller than a specific value (0.01 m in this study), the corresponding virtual observation noise is reduced by a factor of 100.
- 3) Comparison between filtered variance and *a priori* constraint strength, and the number of continuously processed epochs. In an extended Kalman filter (EKF), the parameter variance converges along with the processing. When the filtered variance (predicted variance in EKF) becomes sufficiently small and falls below the calculated constraint strength (observation noise), the STEC corrections have a limited contribution to the estimated state vectors. Furthermore, if an error exists in the correction, the converged solution could possibly degrade due to improper constraining. Therefore, when the parameter has already converged (e.g., when the filtered variance is smaller than the *a priori* constraint strength, or when the

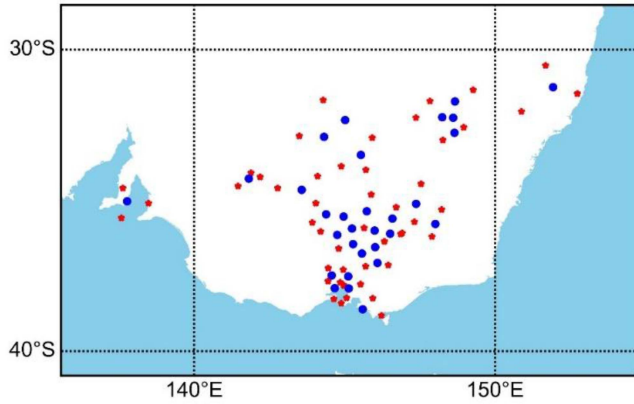


Fig. 1. Distribution of server (red stars) and user (blue dots) stations.

number of continuously processed epochs is larger than 60), the constraining strength is loosened by a factor of ten. Related analysis can be found in Section IV-C.

The thresholds used in this study are empirically set and partly based on the analysis shown in Section IV. The thresholds may change with different interstation distances and solar activities. For instance, the server correction consistency is higher for smaller regions than for larger regions. Therefore, for longer interstation distances, the constraining strength could be loosened to accommodate the differences in signal propagation paths. For stations in low-latitude regions, the thresholds used for checking the consistency can be tightened to avoid the possible influence of ionospheric activities.

III. MEASUREMENT CAMPAIGN

The geophysical network project is established by Geoscience Australia, in which a state-of-the-art network of stations and sophisticated instrumentation that monitors natural and anthropogenic hazards mainly in the Asia-Pacific areas is maintained. As shown in Fig. 1, 30 stations (marked as blue dots), which are capable of tracking GPS and Galileo satellite signals, were selected for positioning validation, with each user station having three nearby server stations (marked as red stars). A total of ten days' datasets, spanning from day of year (DOY) 070-079 in 2023, were collected and processed. Postprocessed PPP AR was conducted for each server station to generate STEC corrections, and simulated real-time PPP-RTK processing was carried out in user stations.

Data from 30 stations are used for user validation, and the one-day observation period was divided into 3-hour batches, resulting in a total of 2400 dataset batches. With the maximum interstation distance of 99.6 km and minimum distance of 30.7 km, the average interstation distance of all stations is 56.6 km. Only GPS and Galileo dual-frequency signals were processed, with the detailed processing strategies outlined in Table I. Different schemes were designed to evaluate the positioning performance based on the number of constrained STEC corrections,

TABLE I
Detailed Processing Strategies in User Stations

Item	Processing Strategy
Frequency	GPS L1/L2, GAL E1/E5a
Station Coordinates	White noise (Kinematic), Constant (Static)
Satellite orbit and clock	Precise products from CODE
Ambiguity	Float, estimated as constant if no cycle slip detected
Ionospheric effect	Estimated as white noise
ZWD	Estimated as random walk parameter
ZHD	Saastamoinen model [35]
Troposphere mapping function	VMF1 [36]

abbreviated IonCst1, IonCst2, ..., and IonCstAll, respectively. The numbers in the abbreviations represent the number of corrections applied, e.g., IonCst1 scheme denotes that only 1 between-satellite differenced slant ionospheric correction is constrained. The float solution and the proposed partial tight-constraining algorithm are abbreviated as Float and IonCstAll-apt, respectively. Moreover, since satellite-differenced STECs are applied as virtual observations within each constellation, the number of constraints is less than the number of available satellite corrections.

IV. RESULTS AND ANALYSIS WITH PROPOSED PARTIAL TIGHT-CONSTRAINING METHOD

The characteristics of the STEC corrections are analyzed first, including the correlation between interpolated STEC errors and satellite elevation angles, the relationship between constraining residuals and elevation angles, the variation of the estimated variance of STEC parameters during processing, and the comparison between constraining strength and filtered variances. Positioning errors, as well as the convergence time, are assessed to evaluate the effect of the number of STEC corrections on positioning performance. Moreover, the efficiency of the proposed partial tight-constraining algorithm is validated.

A. R Correlation Between Interpolated STEC Errors and Elevation Angles

In this contribution, STEC corrections for a user station are calculated using STEC observations from three nearby server stations. Meanwhile, postprocessed PPP AR STEC estimations at the same user station are used as a reference. Hence, the interpolated errors of STEC corrections can be derived and analyzed. Fig. 2 shows the correlation between STEC interpolation errors and elevation angles for all satellites, with different colors representing different satellites. It is obvious that for satellites with low elevation angles, the interpolation errors can occasionally exceed 5 m. Improperly constraining these STEC corrections in PPP-RTK user algorithms may degrade positioning performance. Moreover, significant correlations between interpolation errors and elevation angles are observed for

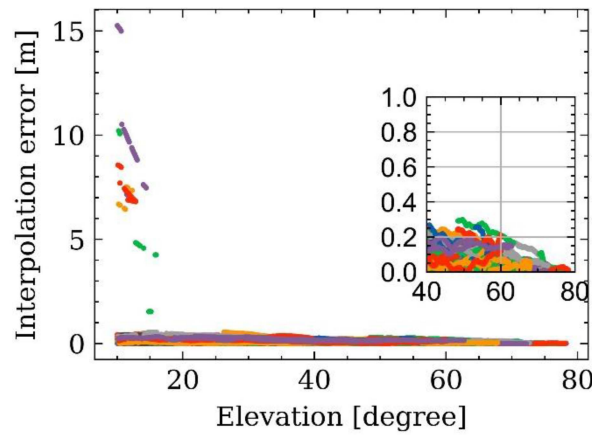


Fig. 2. Correlation between STEC interpolation errors and elevation angles for all satellites at station BLRN on DOY 070 in 2023.

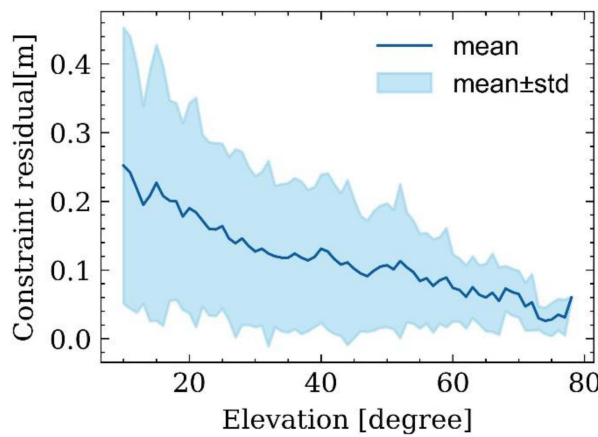


Fig. 3. Correlation between absolute STEC constraint residuals and elevation angles for all satellites at station BLRN on DOY 070 in 2023.

some satellites in the zoom-in plot even though the elevation angle is high, showing that measurements from high-elevation satellites generally exhibit less interpolation errors.

Fig. 3 shows the correlation between absolute STEC constraint residuals and elevation angles, serving as an indication of the dispersion of filtered estimates and corrections. The absolute STEC constraint residuals are the absolute values between the estimated satellite-differenced STECs in the filter and the interpolated satellite-differenced STEC corrections from server stations. The blue line refers to the mean value of the absolute STEC constraint residuals with respect to a specific elevation angle, whereas the edges of the sky-blue shadows represent the sum or subtraction of the mean values and their corresponding standard deviations. With the rise of the satellite, the constraint residuals, as well as the standard deviations, significantly reduce, inferring that the consistency between STEC corrections and estimates is high when the elevation angle is high. These results provide a reference for constraining STEC corrections based on the sequence of satellite elevation angles.

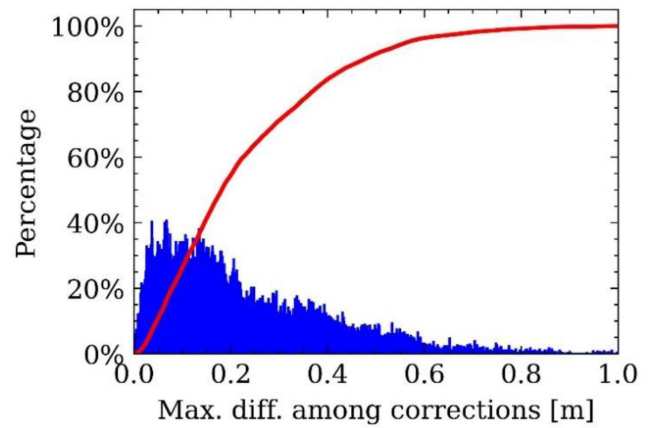


Fig. 4. Distribution and corresponding CDF of the maximum differences among STEC corrections from 3 server stations.

B. Distribution of Maximum Differences Among Corrections From Nearby Server Stations

Normally, STEC observations for the same satellite should be close when the propagation path is similar. For instance, when the stations are within a small region, or when the satellite elevation angle is high. In the PPP-RTK user implementation, a STEC correction consistency check can be performed. The maximum differences of STEC corrections from three nearby server stations are calculated, and the distribution, as well as the cumulative distribution function (CDF), is shown in Fig. 4. Most of the maximum differences are at the decimeter level, with approximately 50% of the differences within 2 dm. The results in Fig. 4 suggest that when correction consistency is high, a higher constraint can be applied.

C. Correlation Between Filtered Standard Deviation of STEC Parameters and Number of Continuously Observed Epochs

The estimates, along with the corresponding variances, converge in the EKF during processing. For a continuously observed satellite without signal blockage or cycle slips, the standard deviation of the ionospheric parameter output from the EKF is expected to remain small enough after a long time. Fig. 5 shows the correlation between estimated STEC standard deviations and the number of continuously observed epochs for all satellites, and different colors represent variable satellites. Noticing that the largest epoch number is 360, as the filter is reset every 3 h. A horizontal line at 0.2 m is also provided as a reference for the constant constraining standard deviation of STEC corrections. It is clear that after 60 epochs, a large number of filtered standard deviations are below 0.2 m, and after ~ 120 epochs, almost all standard deviations are within 0.2 m. The results in Fig. 5 infer that the constraint of STEC corrections can be loosened or removed once the corresponding STEC parameter has converged.

Fig. 6 provides a specific comparison between estimated standard deviations and the constraints for satellite E12

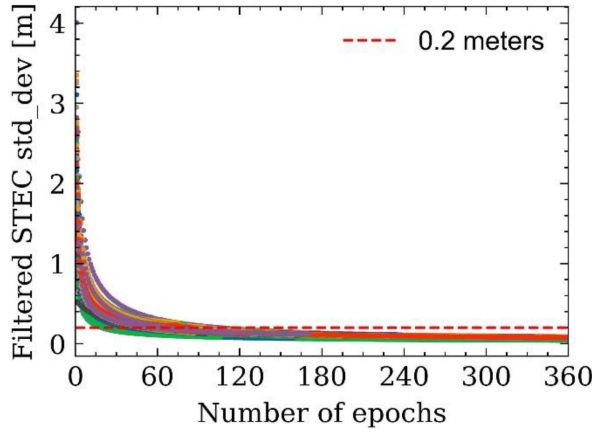


Fig. 5. Correlation between estimated standard deviations and number of continuously observed epochs for all satellites.

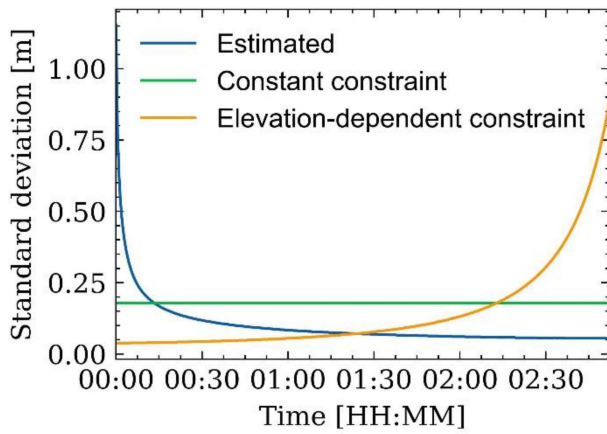


Fig. 6. Time series of estimated standard deviation, constant constraint, and elevation-dependent constraint for satellite E12.

during 00:00–03:00. Both constant and elevation-dependent constraints are illustrated. It can be noticed that with the processing procedure, the estimated standard deviations converge and become smaller than either of the applied constraints. The results indicate that the STEC corrections contribute mostly to the initial stage, i.e., after the convergence of STEC parameters, external constraints play a less significant role in enhancing the positioning solution. Furthermore, the results evolve the strategy of adjusting the constraints according to the estimated standard deviation and the originally applied constraints. Loosening or removing the constraints can either mitigate issues caused by unreliable corrections from the server or reduce the computational burden on the user.

D. Static Positioning Results

Fig. 7 shows the positioning time series and the corresponding number of constrained STEC corrections for a specific station YIEL on DOY 077, 2023. For clarity, only four schemes are plotted in different colors. The positioning errors are notable because of the parameter reset every 3 h. The total number of applied STEC corrections keeps

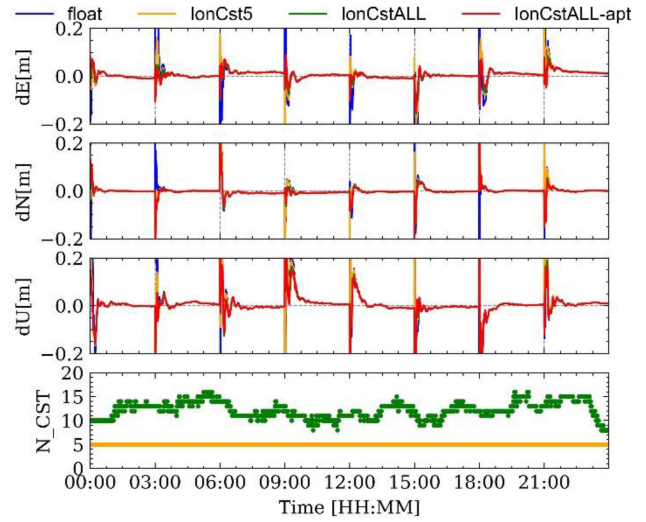


Fig. 7. Static positioning time series of different schemes for station YIEL on DOY 077, 2023.

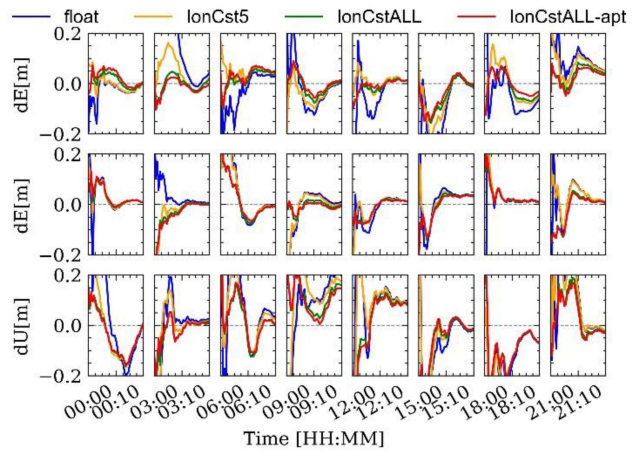


Fig. 8. Static positioning time series for the first 20 min of each reinitialization period of different schemes for station YIEL on DOY 077, 2023.

changing, and there is an average of ~ 12 corrections. To provide a clear view of how the applied STEC corrections contribute to the convergence, Fig. 8 provides a zoom-in plot for the first 20 min of each reset period. Incorporating STEC corrections reduces positioning errors during the convergence. As the number of constraints increases, large errors can be partly mitigated. The proposed adaptive constraint has the best performance, e.g., tighter constraints on part of reliable corrections further reduce the positioning errors compared to conventional methods. For instance, during the convergence time from 03:00, 09:00, and 18:00, the STEC-ALL-apt significantly outperforms STEC-ALL. The constraints play a less significant role after convergence, e.g., minimal differences are observed in the time series after tens of minutes, which is as expected.

The CDF plots of positioning errors for all processed data in the east (E), north (N), and up (U) directions are shown in Fig. 9. The results demonstrate that the proposed

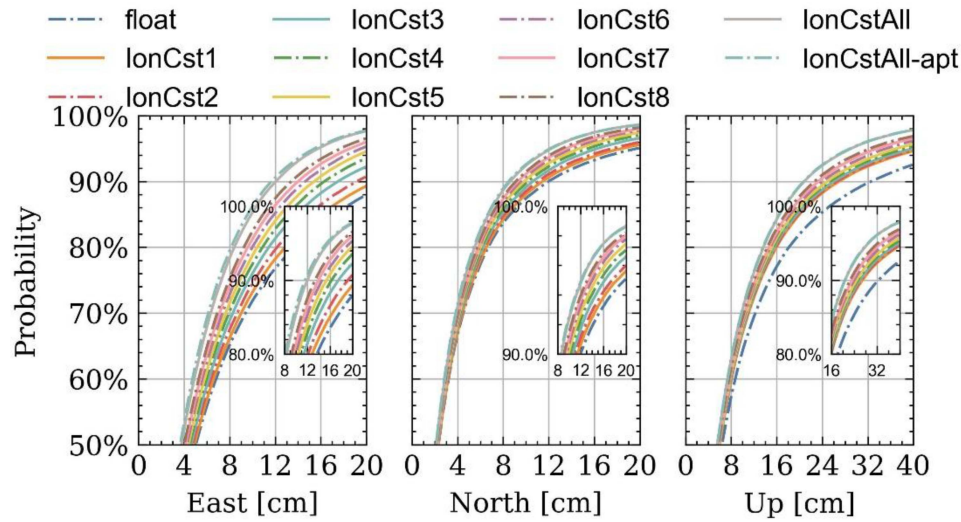


Fig. 9. CDF of static positioning errors for different schemes in E, N, and U, respectively, with all processed data.

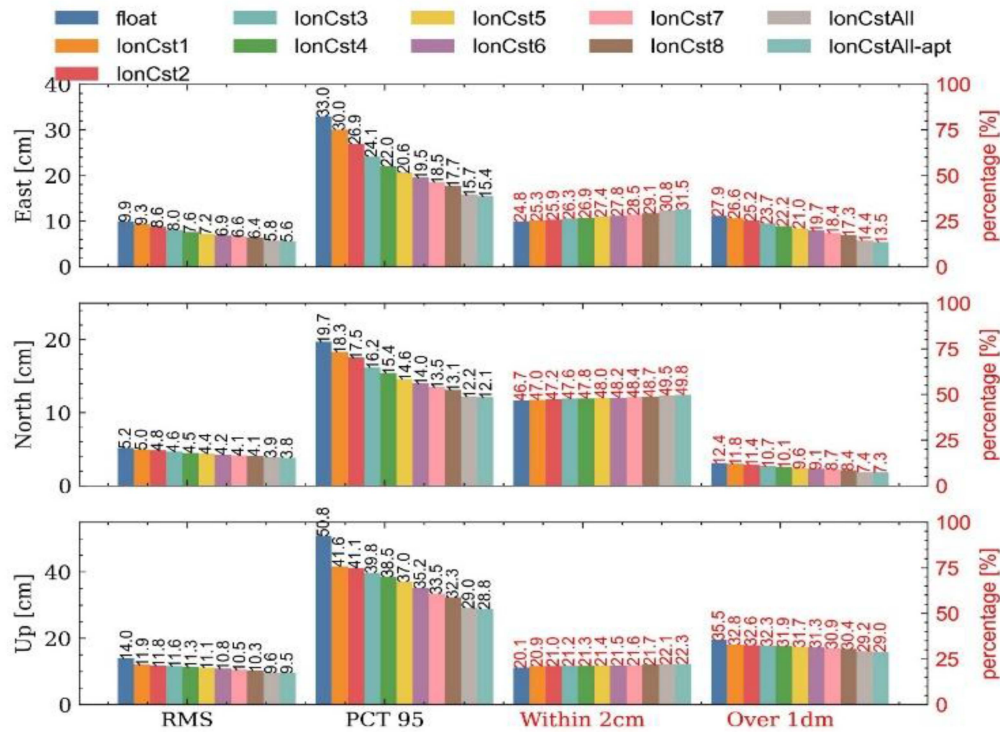


Fig. 10. Statistics of static positioning errors for different schemes in E, N, and U, respectively, with all processed data.

IonCstAll-apt method outperforms other schemes. The most obvious differences are noticed in the east component, and the proposed method achieves a substantially higher cumulative probability within the sub-10 cm error range, indicating superior accuracy and stability. Overall, tighter constraints on reliable corrections provide improvements in accuracy, reliability, and stability, showcasing its potential for practical applications where high-precision positioning is critical.

Four indicators are selected in evaluating the positioning performance of different schemes: the root-mean-square

(RMS) of positioning errors for the first hour, the 95th percentile (PCT 95) positioning errors, and the percentages of errors within 2 cm and over 1 dm, respectively. The RMS shows the performance during the convergence period, PCT 95 provides insight into the large positioning errors, and the percentages indicate solution stability and accuracy. Fig. 10 summarizes the static positioning statistics for all data in the E, N, and U directions, respectively. Notably, even when only one STEC correction is applied, the positioning performance can be improved compared to the float solution, with the most significant improvement observed in PCT 95

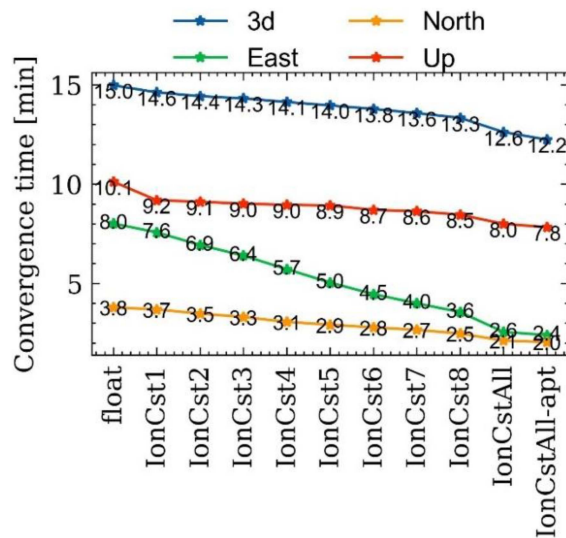


Fig. 11. Convergence time of static positioning errors for different schemes with all processed data.

for the up direction, representing a reduced positioning error of 9.2 cm. As the number of constrained STEC corrections increases, the positioning performance is further improved as well. The proposed adaptive constraining strategy has the best performance across all indicators and directions, representing improvements of 43.4%, 26.9%, and 32.1% in positioning RMS for E, N, and U, respectively, compared to the solution without constraint.

In addition to positioning errors, fast convergence is also a vital indicator of positioning performance. Therefore, Fig. 11 shows the convergence time for three directions, as well as the three-dimensional results for all schemes. The STEC constraints contribute less to the north and up directions, while they significantly shorten the convergence time in the east component. Constraining all corrections can shorten the convergence time by 5.4, 1.7, and 2.1 min in E, N, and U, respectively, compared to scenarios without constraints. By adapting the constraints for selected corrections (IonCstAll-apt), the convergence time can be further reduced by 7.7%, 4.8%, and 2.5% for E, N, and U, respectively.

E. Simulated Kinematic Positioning Results

In the simulated kinematic mode, the positioning errors after convergence are less smooth compared to static mode. However, during the initialization stages, similar trends are observed as in static mode. The simulated kinematic results for the same station are shown in Fig. 12, and the zoom-in plot is shown in Fig. 13, demonstrating that less time is required for the proposed strategy to reach high-accuracy solutions, and the large positioning errors are reduced as well.

The statistics for all processed data are calculated and illustrated in Figs. 14 and 15. The differences are more significant for both east and up components compared to

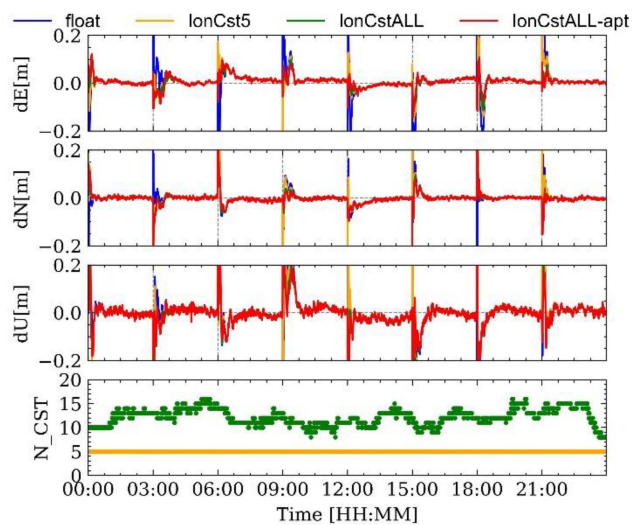


Fig. 12. Simulated kinematic positioning time series of different schemes for station YIEL on DOY 077, 2023.

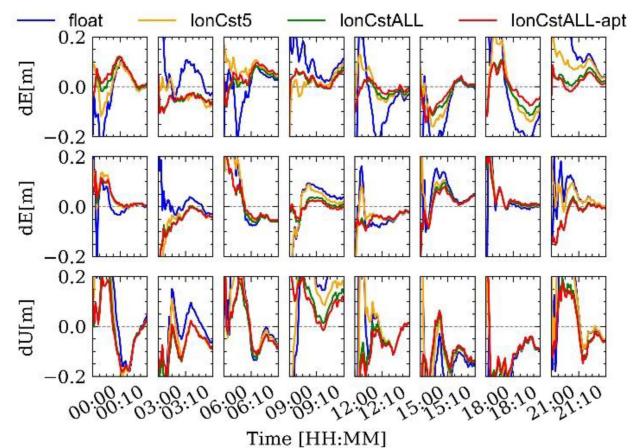


Fig. 13. Simulated kinematic positioning time series for the first 20 min of each reinitialization period of different schemes for station YIEL on DOY 077, 2023.

the north direction, with the proposed strategy consistently delivering the best performance among all schemes. There are decreasing trends in positioning error RMS, PCT 95, and the percentage of positioning errors over 1 dm when the number of STEC constraints increases. Simultaneously, an increasing trend in the percentage of positioning errors within 2 cm is observed as well. The statistics demonstrate that the more the STEC constraints are applied, the better the positioning performance. Besides, by applying the adaptive constraining strength strategy, the large positioning errors can be further reduced, and the best positioning performance can be obtained. Although the positioning performance can be improved with more corrections, when only focusing on the positioning RMS, with extra STEC corrections, the improvement is limited once a certain number of corrections have been applied, e.g., more than 5, especially in the north direction.

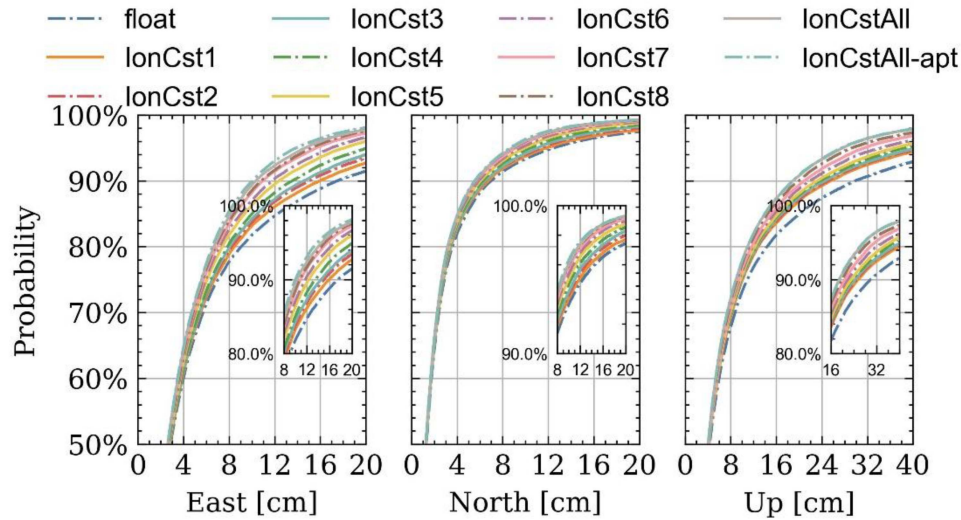


Fig. 14. CDF of simulated kinematic positioning errors for different schemes in E, N, U, respectively, with all processed data.

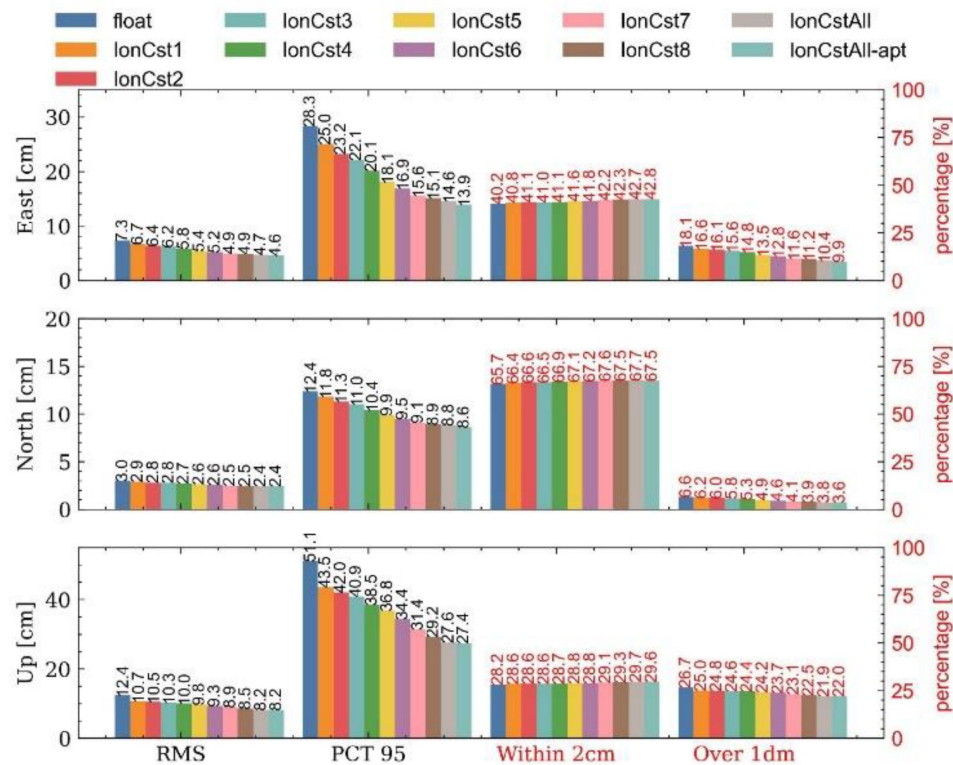


Fig. 15. Statistics of simulated kinematic positioning errors for different schemes in E, N, U, respectively, with all processed data.

Convergence time analysis is also performed in simulated kinematic experiments, as in Fig. 16. Similar to results in static mode, the most significant improvement is noticed in the east component. When only one STEC correction is applied, the convergence time can be reduced by 0.9, 0.3, and 0.9 min in E, N, and U, respectively. By adapting the constraints of partial reliable corrections, the convergence time is the least among all schemes, and can converge within 3.8 and 2.2 min in E and N components, respectively.

F. Real-World Kinematic Experiment

To validate the practical performance of the proposed partial tight-constraining algorithm, a real-world kinematic test was carried out on the playground of Chang'an University, Xi'an, China, on DOY 82 in 2025. The data collected duration was from UTC 02:30 to 03:40 (a total of 70 min), with a sampling rate of 1 s. As shown in Fig. 17, a G970IIPRO receiver (yellow) is equipped on the car, and the receiver on the left (white) is integrated with the Inertial Navigation System (INS) to generate reference trajectories

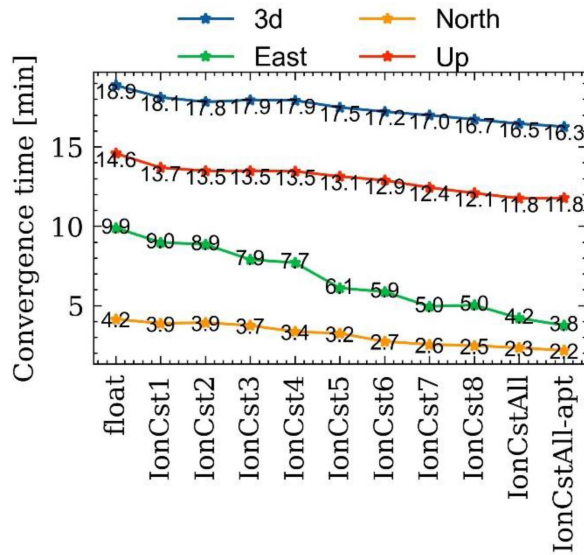


Fig. 16. Convergence time of simulated kinematic positioning errors for different schemes with all processed data.



Fig. 17. Real-world experiment equipment: A G970IIIPRO receiver (yellow) serves as the rover station, whereas the other receiver integrated with INS is used to generate the reference trajectory.

TABLE II
Positioning Performance for the Real-World Kinematic Test

	Positioning Error rms (meter)			Convergence time (second)		
	E	N	U	E	N	U
Float	0.41	0.26	0.16	122	107	50
IonCstALL	0.23	0.14	0.11	110	48	27
IonCstALL-apt	0.21	0.13	0.10	100	45	21

from RTK/INS solutions. Three Continuous Operational Reference System (CORS) stations around the area were used as server stations, with an interstation distance of 78.6 km. Since the proposed algorithm mainly improves the performance during convergence, the filter was reset every 10 min.

Fig. 18 shows the positioning error time series of the east, north, and up directions, respectively, and Table II

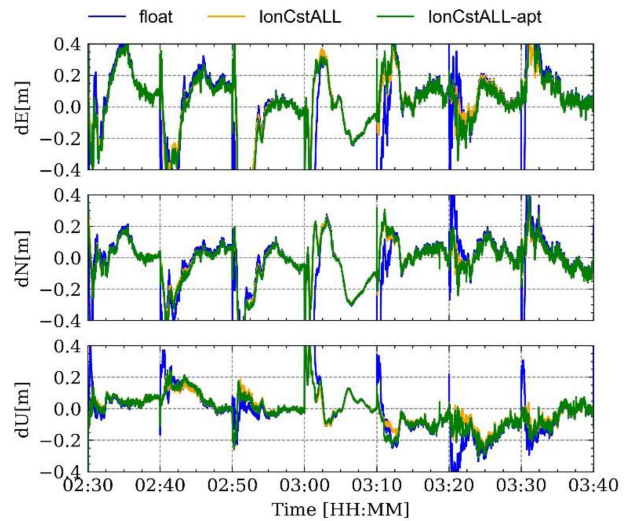


Fig. 18. Time series of positioning errors for the east, north, and up directions for the car experiment.

summarizes the positioning performance including positioning error RMS and convergence time. As the filter was reset every 10 min, the average values of 7 ten-minute periods are calculated. With all corrections applied, the IonCstAll solutions improve the positioning accuracy by 0.18, 0.12, and 0.05 m for east, north, and up directions, respectively, compared to float solutions. With the proposed method, the positioning accuracy is further improved, and the convergence time is shortened by 10, 3, and 6 seconds compared to IonCstAll solution. Despite the interstation distance in this test is relatively long, when correctly diagnosing the reliable STEC corrections and applying tight constraints, the solutions get better than those obtained from conventional algorithm.

V. CONCLUDING REMARKS

With STEC estimates from three nearby server stations, the corresponding correction can be calculated and constrained at user stations. Based on evaluation of the effect of number of corrections on positioning performance, this contribution proposes an improved constraining algorithm in which tight constraints are applied in the selected reliable corrections, aiming to further enhance the positioning performance. The proposed algorithm is validated with both static and simulated kinematic positioning results.

The characteristics of the interpolated STEC corrections are first analyzed. Both the interpolation errors and constraining residuals show significant correlations with satellite elevation angles, which drives the conducted experiments in constraining STEC corrections in an elevation angle manner. The relationship between estimated variance of STEC parameters and number of continuously processed epochs are assessed, which further motivates the analysis in comparison of filtered variance and applied constraint strength. The results show that, after convergence, the estimated variance itself is small enough, and can be smaller than the applied constraints after a long time, which partly

demonstrates the correctness of the proposed strategy in which the constraint is loosened after a long time.

Both static and simulated kinematic positioning results show that the positioning performance will be better when applying more STEC corrections, which is as expected. The most significant improvement is noticed in the east direction, while a limited contribution is observed when the number of STEC corrections is already larger than 5. When tight constraints are applied in several reliable corrections, the static positioning errors can be reduced by 43.4, 26.9, and 32.1% in positioning RMS for E, N, and U, respectively, compared to the solution without constraints. In simulated kinematic results, the convergence time can be further shortened by 9.5 and 4.3% for E and N, respectively, compared to those when all corrections are applied.

The interstation distances among the stations used in this study range from ~ 30 km to ~ 100 km, and the overall statistics demonstrate that the proposed method works well across different interstation distances. Moreover, since the basic principle of the proposed method is to validate the reliability of the corrections obtained from nearby server stations, such correction consistency checks are also expected to work well for stations in low-latitude regions where the GNSS observations are frequently affected by ionospheric activities such as scintillation and ionospheric bubbles.

The results presented in this contribution serve a dual purpose. First, they provide a valuable reference for PPP-RTK servers and users, offering guidance on the number of STEC corrections required to meet the demands of various users. This is particularly beneficial for those with limited computational resources, such as low-cost users. Second, the proposed partial tight-constraining strategy significantly enhances positioning performance compared to conventional methods, delivering more accurate, faster, and more reliable positioning solutions.

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